

Virtual Try-On

Stage 13.9 — Integration Test Report

1. Testing Objective

The purpose of integration testing was to verify the interaction between all application modules and ensure the stability of the Virtual Try-On system under real usage conditions.

The testing focused on:

- correct initialization of all modules;
 - interaction between the camera and Face Detector;
 - accessory rendering pipeline;
 - photo saving;
 - video recording;
 - complete user workflow;
 - long-term application stability.
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2. Test Environment

Software

Component	Version
Operating System	Windows 10 Pro 22H2 (64-bit)
Python	3.12
OpenCV	4.11.0
MediaPipe	0.10.x
NumPy	2.x
Pillow	11.x
CustomTkinter	5.x

Hardware

Component	Specification
CPU	AMD Ryzen 5 4500U
RAM	16 GB
GPU	AMD Radeon Graphics
Camera	Integrated HD Webcam

3. Test Scenarios

The following integration scenarios were executed.

Stage 13.1

Application Startup

- application launch
 - main window creation
 - module initialization
 - exception handling
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Stage 13.2

Camera Integration

- camera opening
 - frame acquisition
 - frame transmission
 - preview update
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Stage 13.3

Face Detection Pipeline

- camera
 - face detection
 - landmark extraction
 - renderer communication
-

Stage 13.4

Accessory Rendering Pipeline

- category selection
 - accessory selection
 - PNG loading
 - rendering
 - preview update
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Stage 13.5

Photo Saving Integration

- rendered frame
 - photo saving
 - file creation
-

Stage 13.6

Video Recording Integration

- recording initialization
 - frame writing
 - recording stop
 - video creation
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Stage 13.7

Full User Scenario

- application launch
- camera start
- face detection
- accessory selection

- preview
 - photo saving
 - video recording
 - application shutdown
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Stage 13.8

Stability Testing

- repeated camera restart
 - fast accessory switching
 - cache cleanup
 - resource release
 - long-running execution
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4. Test Results

Test	Result
Application Startup	✓ Passed
Camera Integration	✓ Passed
Face Detection Pipeline	✓ Passed
Accessory Rendering Pipeline	✓ Passed
Photo Saving Integration	✓ Passed
Video Recording Integration	✓ Passed
Full User Scenario	✓ Passed
Stability Testing	✓ Passed

5. Issues Found During Testing

Several issues were identified during the development process and successfully resolved.

Issue	Status
Camera initialization	Fixed
Face Mesh detection validation	Fixed
Accessory cache handling	Fixed
Empty frame processing	Fixed
Photo saving validation	Fixed
VideoWriter initialization	Fixed
Renderer alpha blending	Fixed
GUI callback warnings after window destruction	Known limitation of CustomTkinter during automated testing

6. Overall Result

A total of eight integration testing stages were completed successfully.

The Virtual Try-On application demonstrated stable interaction between all software modules, correct accessory rendering, successful multimedia processing, and reliable operation during prolonged execution.

No critical functional defects preventing normal system operation were identified.

7. Conclusion

The integration testing confirmed that all major subsystems operate correctly as a unified software solution.

The application satisfies the functional requirements defined in the Software Requirements Specification (SRS) and is considered ready for deployment and final acceptance testing